University of Nevada, Reno

Smart-Cities Enabled by Multi-radio Multi-channel Heterogeneous Networks

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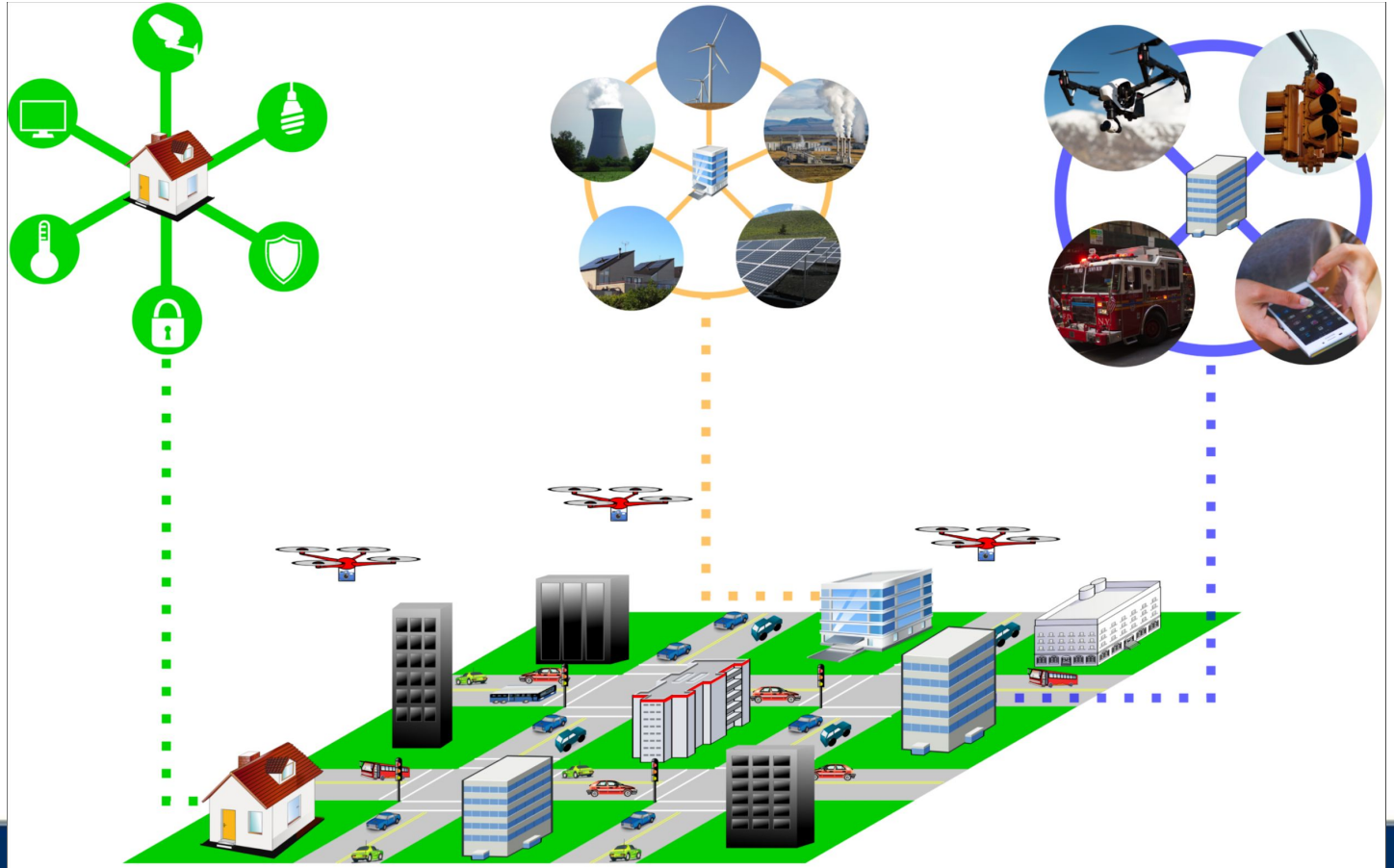
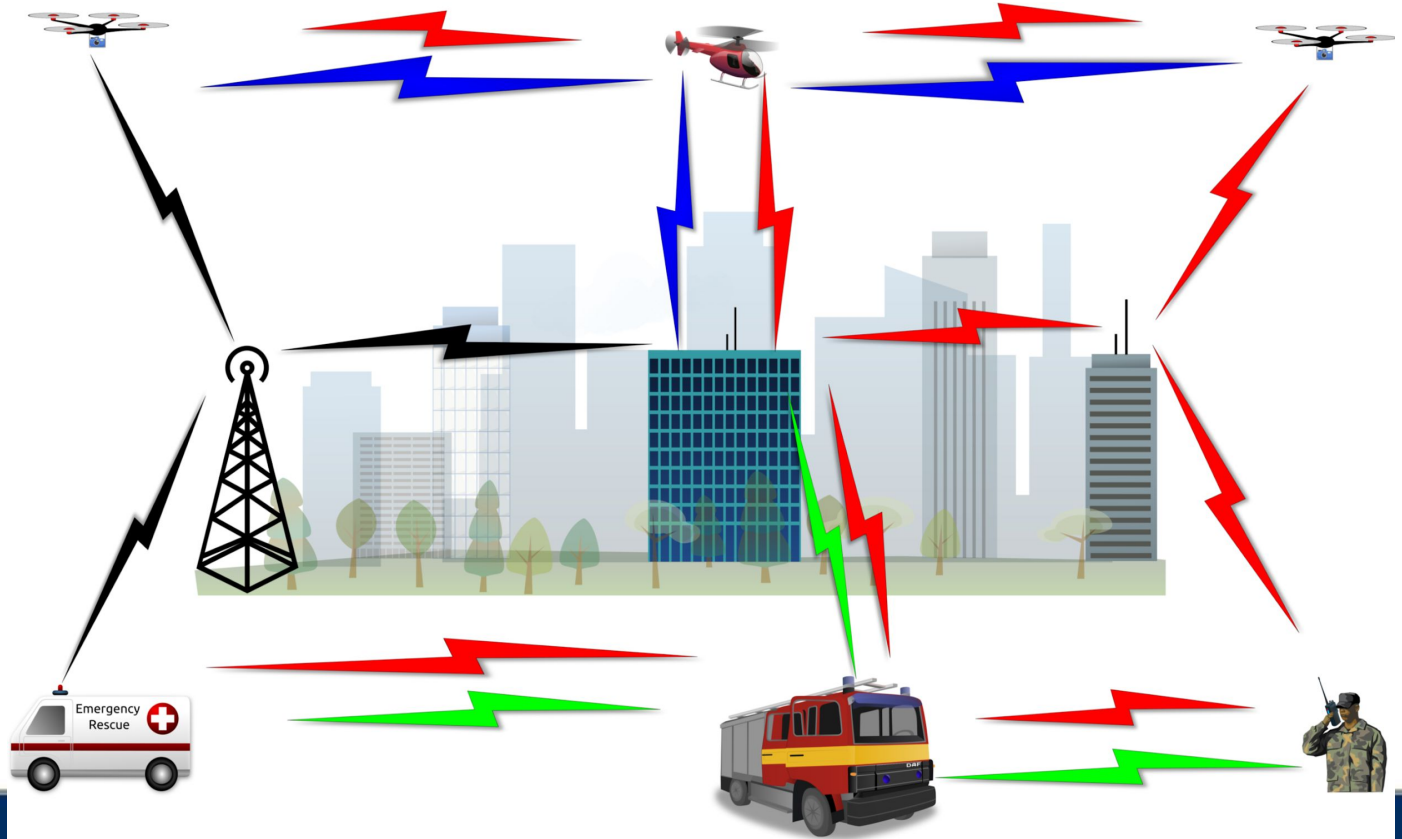
What is a Smart-City?



What is a Smart-City? Why should we care?

- Using data analysis to make better/automated decisions
- Multiple (different) hardware and software technologies to solve each small piece of the puzzle
- Examples:
  - Smart-house and smart-grid
  - Traffic management
  - Temporary service provision

Multi-radio multi-channel HetNets



Multi-radio multi-channel HetNets

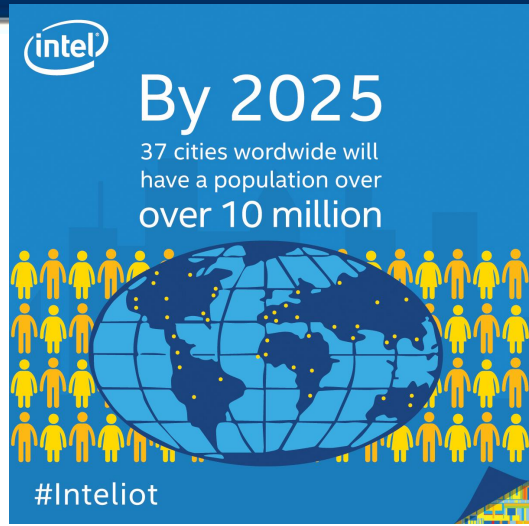
- Multi-radio: more than one transceiver, each capable to transmit in a specific channel
- Multi-channel: ability to transmit in more than a single frequency
- Heterogeneous:
  - Nodes might have different radios available
  - Nodes may move differently
  - Nodes can have different energy source (UAV/cars/base stations)

NSF:S&CC initiative

- NSF: Smart & Connected Communities
  - Funding opportunities
  - Interdisciplinary efforts
  - Broad foreseen impact: agriculture, infrastructure, emergency management, smart electric power grid, health, public safety, rural and urban planning, transportation, water management, etc.

Commercial potential

- Private sector is already deploying solutions[1]
- Improve quality of life generates revenue
- Efficient resource management: smart-grid, water distribution
- Decision/policy making: data gathered and analyzed help legislators



1 - https://newsroom.intel.com/news-releases/san-jose-implements-intel-technology-for-a-smarter-city/

3D obstacle compliant mobility model

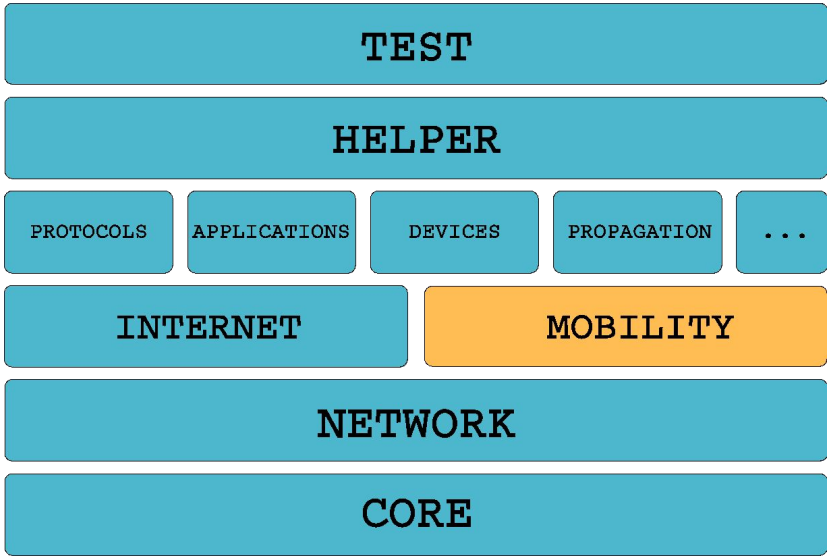
- Network simulator (ns-3) provide vast amount of libraries to allow simulations of different communications scenarios and technologies.
- Mobile Ad Hoc Networks is a possible scenario to simulate.
- Mobility models dictate how nodes move around.
- Physical objects (i.e. buildings) are accounted from a communications perspective but not from a mobility model point-of-view.

Existing Models

- Random Walk (2D):
  - Move in a straight line for a determined amount of time.
  - Randomly choose new parameters.
- Random Direction (2D):
  - A node moves in straight line until it reaches the boundary.
  - Randomly chooses new parameters.
- Gauss-Markov (3D):
  - Moves for a certain time before changing parameters.
  - Parameters can be constrained by random distributed variables.

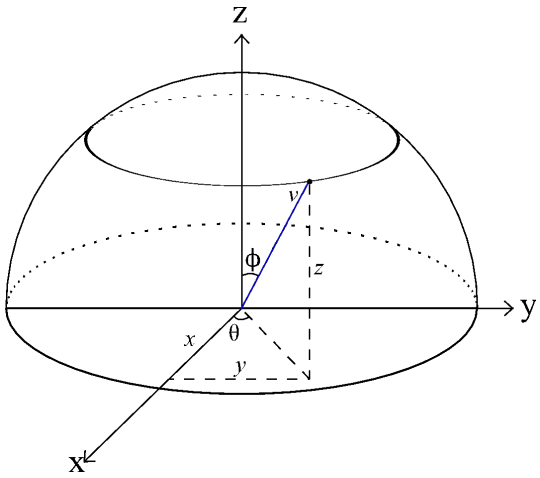
Scope of work

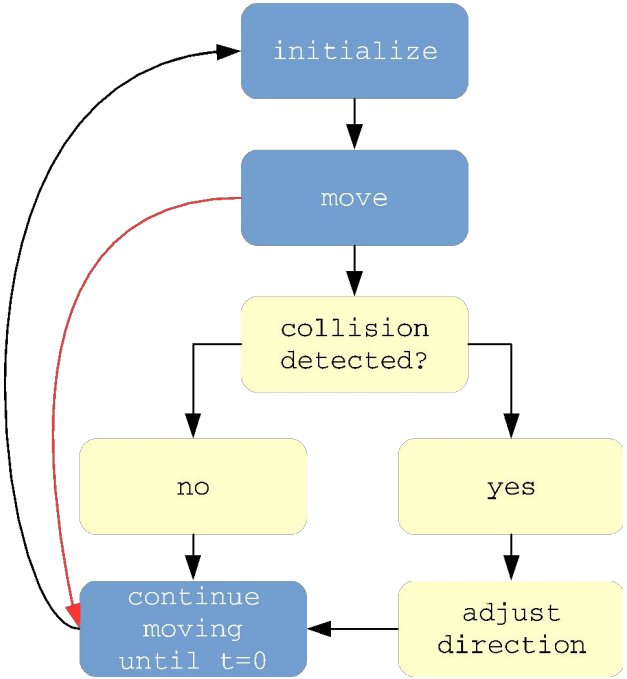
- New classes added to the mobility module
- Modification of existing auxiliary classes
  - No functions were removed/modified
  - Only new methods added to account for 3D geometry



General block diagram

- All models follow similar flow
- Intermediate yellow blocks represent the new features
- Coordinate system used in ns-3





Random Walk (distance mode)

Random Walk (time mode)

Random Direction

Gauss-Markov

External links

- Project website:

http://www.cse.unr.edu/~regis/ns-3/

- Source-code on GitHub:

https://github.com/regisin/ns-3-obstacles

Future work

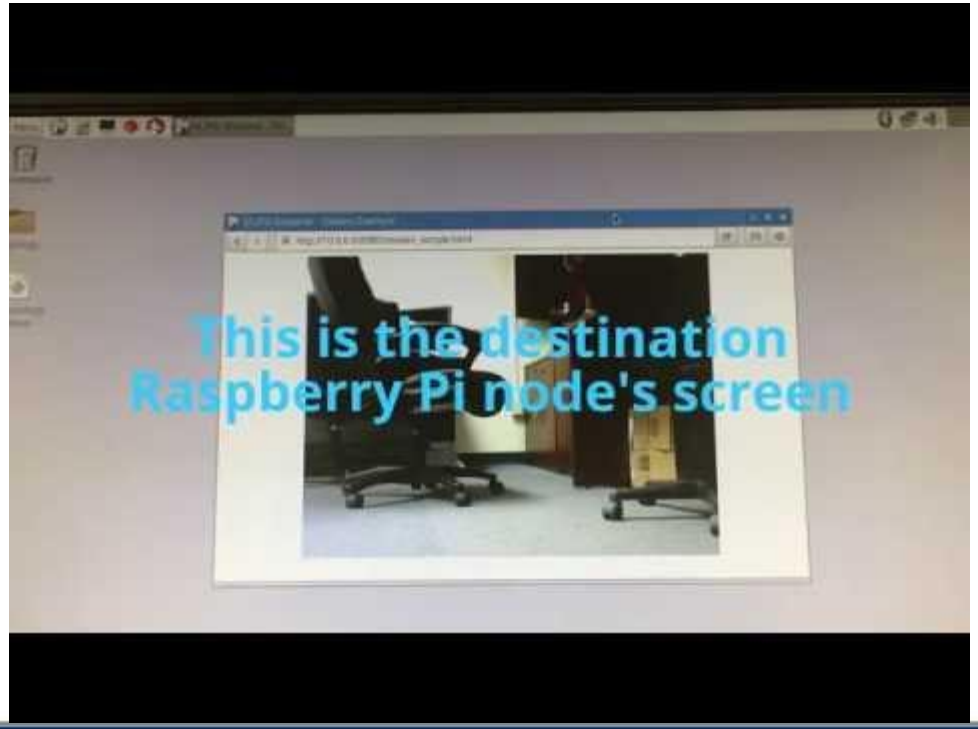
- Network-aware application and application-aware network:
  - Application will adapt depending on current network conditions (generate data in a different if network is lossy)
  - Network will create redundant packets and send them through different paths to reduce risk of loss

Future work

- Network-constrained node deployment/placement:
  - Find the best positions to deploy a UAV network without compromising connectivity and increasing coverage
  - Minimize the number of aircrafts needed, saving resources and potential for interference

Future work

- Testbed development (in parallel):
  - Build a proof-of-concept testbed where the proposed/future mechanisms can be validated
  - Have modularity in mind to be able to update the hardware as new technologies become available



Thank You!